

Oracle Database 12c Complete Course Note

Introduction

Oracle Database 12c is a powerful relational database management system designed for enterprise-level applications. It supports legacy systems while introducing modern features like multitenant architecture and cloud readiness.

Chapter 1: Architecture

Database Instance

An Oracle instance consists of memory structures and background processes. It manages database operations and user connections. Each instance is associated with one database.

Database Storage

Oracle databases use physical files such as datafiles, control files, redo logs, and archive logs. These files store actual data and metadata required for database operation.

Server Processes

Server processes handle user requests, execute SQL statements, and manage data retrieval and manipulation. They work alongside background processes to ensure efficient database functioning.

Chapter 2: Instance Memory Pools

System Global Area (SGA)

SGA is a shared memory region that contains data and control information for one Oracle instance. It includes several subcomponents.

Shared Pool

Stores parsed SQL statements, PL/SQL code, and data dictionary information. It improves performance by reusing execution plans.

Buffer Cache

Caches frequently accessed data blocks to reduce disk I/O and improve performance.

Redo Log Buffer

Temporarily stores redo entries before writing them to redo log files. Essential for recovery operations.

Other SGA Pools and Program Global Area (PGA)

Includes Java Pool, Large Pool, and Streams Pool. PGA is used for session-specific memory, such as sort operations and session variables.

SGA and PGA Sizing

Proper sizing is critical for performance. Parameters like SGA_TARGET and PGA_AGGREGATE_TARGET are used to configure memory allocation.

Chapter 3: Instance Background Processes

Database Writer (DBWn)

Writes modified blocks from the buffer cache to datafiles.

Log Writer (LGWR)

Writes redo log buffer contents to redo log files.

Checkpoint Process (CKPT)

Updates datafile headers and control files during checkpoints.

SMON (System Monitor)

Performs instance recovery and coalesces free space.

PMON (Process Monitor)

Cleans up failed user processes and releases resources.

RECO (Recoverer)

Handles recovery of distributed transactions.

LREG (Listener Registration)

Registers the instance with the listener for client connections.

Archiver (ARCn)

Archives filled redo log files for backup and recovery.

Chapter 4: Client Connections**Oracle Listener**

A server-side process that listens for client connection requests and establishes communication with the database instance.

TNSNAMES.ORA Configuration Files

Contain network service names mapped to connect descriptors. Used by clients to locate and connect to databases.

How Client Connections Work

Clients use Oracle Net to communicate with the listener, which then connects them to the appropriate server process.

Chapter 5: Physical Database Structures**Database Storage File Types**

Includes datafiles, control files, redo log files, archive logs, and temporary files.

Datafiles and Tablespaces

Datafiles store actual data. Tablespaces are logical storage units that group datafiles.

Special Oracle Tablespaces

Includes SYSTEM, SYSAUX, UNDO, TEMP, and USERS tablespaces, each serving specific roles.

Redo Log Files

Record all changes made to the database. Used for recovery in case of failure.

Control Files

Contain metadata about the database structure, including datafile locations and log history.

Backup Files

Created using RMAN or user-managed methods. Essential for disaster recovery.

Additional Storage Files

Includes archived redo logs, flashback logs, and dump files.

Chapter 6: Multitenant Database

Benefits of the 12c Architecture

Supports pluggable databases (PDBs) within a container database (CDB), enabling better resource management and isolation.

New vs. Old

Traditional architecture used single databases. 12c introduces multitenancy for cloud and enterprise scalability.

Backward Compatibility

Supports older applications and configurations, ensuring smooth transition to 12c.

Chapter 7: Start and Stop the Database

Startup

Involves three stages: NOMOUNT (initialization), MOUNT (control file access), and OPEN (full database access).

Shutdown

Can be performed in NORMAL, IMMEDIATE, TRANSACTIONAL, or ABORT modes depending on the situation.

Chapter 8: Install Oracle 12c Software

Prerequisites for Installation

Includes hardware requirements, OS compatibility, and necessary user permissions.

Operating System Configuration

Involves setting kernel parameters, creating Oracle user and groups, and configuring environment variables.

Download the Software

Available from Oracle's official website. Requires Oracle account for access.

Install Software

Performed using Oracle Universal Installer (OUI). Includes selecting installation type, configuring paths, and verifying prerequisites.

Chapter 9: Create a Database Instance

Use NETCA to Create a Listener

Network Configuration Assistant helps configure and start the Oracle listener.

Use DBCA to Create a Database

Database Configuration Assistant guides through database creation, including naming, memory allocation, and storage configuration.

Chapter 10: Tools for Developers and DBAs

SQL*Plus for Administration

Command-line tool for executing SQL commands and managing the database.

SQL Developer for Development

Graphical interface for writing and executing SQL and PL/SQL code. Supports debugging and data modeling.

Oracle Enterprise Manager

Web-based tool for monitoring, managing, and tuning Oracle databases.

Chapter 11: Management

Start and Stop the Instance

Use SQL*Plus or Enterprise Manager to manage instance lifecycle.

Start and Stop the Listener

Use the `lsnrctl` utility to control the listener process.

View and Set Parameters

Use `SHOW PARAMETER` and `ALTER SYSTEM` commands to manage initialization parameters.

Create Users and Assign Permissions

Use `CREATE USER` and `GRANT` statements to manage user access and roles.

Data Dictionary

System tables that store metadata about database objects, users, and configurations.

Troubleshoot Using the Alert Log

Contains messages and errors generated by the database. Located in the diagnostic directory.

Storage: Tablespaces and Datafiles

Manage storage by creating, resizing, and monitoring tablespaces and associated datafiles.